Activation-Side Session Tagging for LLM Audit Logs: What a Residual-Stream Watermark Can and Cannot Prove

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Abstract

We study session tags on logged transformer activations. Residual perturbations are recovered by subtracting a clean replay of the disputed text. Each condition uses 128 sessions, 128 generated tokens and injection fraction 0.01. At fixed 0.5B, fp32 to bf16 lowers identical continuations from 0.688 to 0.258 and paired accuracy from 0.978 to 0.904. Precision dominates this measured contrast; bf16 size effects are smaller and nonmonotonic. Blind calibration remains near chance. Read-layer substitution recovers all sixteen bits in every measured model without the intervening marked computation: decoding does not prove execution. We separate physical-coordinate detection from sixteen-bit payload recovery. Sixty-four-coordinate detection strengthens nominal session-level tails without adding payload entropy. Long-rollout fidelity is substantially worse than short-horizon agreement suggested. Commit-then-verify semantics use externally witnessed Merkle roots. The tag supplements signed transcripts with an internal consistency check, not non-repudiation against its provider or black-box detection. Its incremental utility over signed unmodified activation logs is unproven; our contribution is characterization and negative results. Paired verification requires private access to text.

1 Introduction

LLM audits ask what text was recorded, whether the associated internal trace is consistent, and whether the claimed computation occurred. A signed transcript addresses the first question. It does not prove the truth of a dishonest signer's original statement. An activation tag addresses the second question under capture and custody assumptions. It does not answer the third.

The earlier version [7] claimed legal non-repudiation from a symmetric key and logs controlled by the provider. That claim is withdrawn. The provider can fabricate a matching record or present an unmarked record to seek exculpation. The earlier protocol description also conflated static calibration with clean-state subtraction and reported pooled prompt recovery as one-session evidence.

We replace those claims with an exact paired/blind protocol, session-level statistics, long autoregressive fidelity measurements and null/substitution controls across Qwen configurations. We distinguish physical detection strength from logical payload recovery. Our central negative findings are limited blind recovery, long-horizon text divergence and successful known-key fabrication at the read layer. The remaining role is a session label on a precommitted activation log.

A signed text hash should always accompany that label. It already binds exact text more strongly and cheaply. The additional activation channel can test whether a trace is consistent with a replay of the disputed transcript. The channel is separate from text storage, not statistically independent of text. Compact projections permit commitment without publishing text; they do not eliminate the paired verifier's need to access it.

2 Threat model and commit-then-verify semantics

The provider holds key K, runs inference and retains private text and activations. A client disputes a transcript. An auditor receives authenticated records and controlled access to the model snapshot, key and text. A separately witnessed append-only log or regulator timestamps commitments before a dispute. Merkle inclusion and consistency checks follow the transparency-log pattern [4]. Provider timestamps alone do not prevent retrospective rewriting.

Let x and y be prompt and emitted token sequences. With length-delimited canonical serialization Enc, the proposed inference-time record is

$$q = H(\mathrm{Enc}(x, y)), \qquad z = C_K^\top \bar{h}'_{\mathrm{read}} \in \mathbb{R}^B \quad \text{or} \quad \bar{h}'_{\mathrm{read}} \in \mathbb{R}^d, \tag{1}$$

$$r = \mathrm{Enc}(\mathrm{sid}, \tau, q, \mathrm{model_hash}, \mathrm{protocol_hash}, \mathrm{key_id}, b, T, z), \tag{2}$$

$$u = H(\mathrm{Enc}(r, \rho)), \qquad \mathcal{R}_j = \mathrm{MerkleRoot}(u_1, \ldots, u_j). \tag{3}$$

The salt ρ prevents simple dictionary tests against predictable records. The model hash binds weights, architecture and tokenizer. The protocol hash fixes layers, precision, hooks, code generation, averaging positions, float encoding, threshold, and the blind calibration identifier/hash (corpus, dataset revision, selected prompt/token sequences and calibration-vector serialization). Key version and payload mapping are frozen before disputes. Deployment also needs signatures, retention rules and an audited publication deadline; commitments do not enforce complete capture or prevent selective omission.

The auditor checks inclusion, timing, key/version commitments and transcript hash before evaluating the activation tag. A hash mismatch already detects exact text alteration. For a matching candidate, paired replay tests internal consistency with the committed trace. Exculpation is restricted to timely committed records under a validated logging policy. Neither a missing record nor a nonsignificant tag establishes non-origin.

A full-window projection suffices for the owner paired check because averaging and projection commute. Arbitrary prefix checks or new keys require richer traces or additional precommitted projections. Publishing only the Merkle root conceals these values, but disclosed activations are not guaranteed to hide text. The transparency integration is proposed architecture; the experiments evaluate the tag and do not implement a production commitment service.

Comparison with signed logs. The baseline signs $H(\mathrm{Enc}(\mathrm{sid}, \tau, x, y, \mathrm{model_hash}))$. Ed25519 signatures occupy 64 bytes, excluding the record and Merkle path, and provide approximately 128-bit security [2]. HMAC-SHA-256 normally uses a 32-byte MAC and is symmetric. A SHA-256 fixed-target second-preimage idealization has scale 2^{-256}, versus 2^{-128} for generic collisions. These are cryptographic security scales, not statistical p-values. Signing introduces no model-distribution change: $D_{\mathrm{KL}} = 0$.

An essential ablation is a *signed unmodified activation log*: recording and signing clean activations, or $C_K^\top \bar{h}$ without injection, permits the same replay-consistency comparison with $D_{\mathrm{KL}} = 0$. Perturbation adds a known-payload statistical test and a key-conditioned session label; a signature already provides key binding. Its incremental utility over this unmodified-log baseline is not demonstrated here. The contribution is an empirical characterization, including negative results, rather than a proven audit advantage.

An external commitment prevents undetected later edits, not earlier fabrication. Wrong-key controls represent fixed non-owner decoders. A key-holding provider or worker is stronger: it can attach the code at the read layer without carrying it through the marked intermediate computation. The public research keys are test values, not production secrets.

Property	Signed transcript + commitment	Activation tag + same commitment
Exact text binding	Primary cryptographic mechanism	Still requires the text hash
Internal consistency	Text log alone: no activation comparison	Replay against marked states
Signed unmodified activation log	Clean trace/projection also supports replay; $D_{\mathrm{KL}} = 0$	Known-payload statistical test; incremental utility unproven
Private commitment	Unpublished text can be hashed	Compact projection can be committed
Detection guarantee	Signature/hash assumptions	Statistical null and capture assumptions
Dishonest provider	Can sign a false original claim	Can fabricate before commitment
Proof of execution	No	No; substitution passes

Table 1: The activation tag supplements transcript integrity; it does not strengthen its cryptographic guarantee.

3 Mechanism and exact protocol

3.1 Injection and alignment

For width d and L blocks, zero-indexed output hooks use $\ell_{\mathrm{inj}} = \mathrm{round}(L/3)$ and $\ell_{\mathrm{read}} = \mathrm{round}(5L/6)$. With payload $b \in \{-1, +1\}^B$,

$$C_K \in \{-d^{-1/2}, +d^{-1/2}\}^{d \times B}, \qquad h'_t = h_t + \alpha \|h_t\|_2 \frac{C_K b}{\|C_K b\|_2 + \epsilon}. \tag{4}$$

SHA-256 of K seeds a CPU PyTorch generator for sign-normal code columns. The first sixteen of sixty-four columns define the uncoded condition. This is a reproducible PRNG construction, not a security-hardened PRF.

Greedy generation uses the KV cache. The output hook marks the prompt's final position and each subsequent position predicting an emitted token. Logged h'_t therefore predicts y_t; it is not the state containing y_t. Marked cache states persist. For prompt length P, clean replay reads positions $P-1$ to $P+T-2$. The record stores T prediction states or their declared projection/mean. Experimental payloads are sixteen random signs seeded by $10000 + i$, not unique session identifiers. The full identifier remains in Eq. (2).

3.2 Paired and blind verification

Paired verification disables injection and teacher-forces the exact disputed sequence $x \| y$ to obtain aligned clean states. It does not subtract states from an independently generated clean continuation after its text diverges:

$$s^{\mathrm{paired}} = \frac{1}{T} \sum_t C_K^\top [h'_t - h_t(x, y_{<t})] = \bar{s}' - C_K^\top \bar{h}(x, y), \qquad \hat{b} = \mathrm{sgn}(s). \tag{5}$$

Replay needs no embedding key; decoding still needs K or C_K. The model, tokenizer, precision and hook alignment must match. Numerical differences between cached generation and full teacher forcing are included in the measured residual.

Blind verification instead subtracts static calibration from disjoint clean continuations:

$$c_K = M^{-1} \sum_{u=1}^{M} C_K^\top h_u^{\mathrm{calib}}, \qquad s^{\mathrm{blind}} = \bar{s}' - c_K. \tag{6}$$

This removes corpus-mean bias, not prompt-specific activation variation. Blind accuracy remains near chance across the measured configurations; the audit verifier therefore uses paired replay.

3.3 Physical detection versus payload recovery

At $B = 16$, signs directly encode sixteen payload bits. At $B = 64$, each logical bit is repeated in four distinct code columns. Majority voting recovers the sixteen-bit payload; a two–two tie maps to +1, as do zero scores. Repetition adds redundancy, not payload entropy.

Detection uses physical matches m across $n = B$ code coordinates. Under the independent fair-sign null, its one-sided session tail is

$$p_{\text{phys}}(m; B) = 2^{-B} \sum_{k=m}^{B} \binom{B}{k}. \tag{7}$$

Repeating target signs does not itself correlate independent null code decisions; it is therefore legitimate to test all sixty-four physical coordinates under that null. Their errors can be correlated under the marked alternative without invalidating a null-defined test. Payload recovery is a separate quantity: logical bit accuracy and the optional sixteen-logical-sign reference tail after majority vote. Under independent fair physical signs, tie-to-+1 makes a null logical vote positive with probability 11/16, not 1/2. Logical tails are therefore labelled "nominal (payload-randomised)": the fair-match reference averages over independent random payload signs and is not a conditional fixed-payload guarantee.

For $B = 16$, $p(14; 16) = 2.09 \times 10^{-3}$ and $p(16; 16) = 1.53 \times 10^{-5}$. Only sixteen-coordinate statistics have this floor; sixty-four-coordinate detection can pass 10^{-5}. Fixed code overlaps and

Matches	Nominal one-session tail
12/16	3.84×10^{-2}
13/16	1.06×10^{-2}
14/16	2.09×10^{-3}
15/16	2.59×10^{-4}
16/16	1.53×10^{-5}

Table 2: Exact one-sided binomial examples (not empirical scale results). For $B = 16$, the minimum tail exceeds 10^{-5}; $B = 64$ physical detection is not subject to this floor.

shared activation structure may depart from independent fair null signs. We therefore call the binomial tails nominal and report physical wrong-key/no-mark exceedances. Neither a low tail nor good payload recovery is an adversarial forgery bound. Thresholds are strict and applied per session; deltas are never pooled across prompts. Searching keys, payloads or windows would require a separate multiple-testing policy.

4 Experimental setup

The study covers Qwen2.5-0.5B fp32/bf16, 1.5B bf16 and 7B bf16. All tables are generated from the result JSONs; Table 3 identifies measured conditions. Pending configurations: none.

Each prescribed condition uses 128 sessions, 128 separate calibration prompts, 128 generated tokens, $\alpha = 0.01$, and physical $B \in \{16, 64\}$. From WikiText-2 validation (`Salesforce/wikitext`, `wikitext-2-raw-v1`), the script takes rows of at least 200 characters, excluding headings, with enough tokens. The first eligible rows supply session prompts; the next supply calibration. Prompts use their first 32 tokens. Calibration uses clean generated continuations, not dataset suffixes. Torch seed is zero, dropout is disabled and decoding is greedy.

Fidelity. Clean and marked runs freely generate full continuations. We measure the fraction entirely identical, zero-indexed first divergence (censored at T), and full-sequence token agreement. Conditional $D_{\text{KL}}(p'_t \| p_t)$ is averaged through the first differing prediction inclusive, while token histories still agree. Hidden caches already differ. Subsequent token agreement measures trajectory similarity, not same-context predictive accuracy or semantic quality.

Model/dtype	d	L	ℓ_i	ℓ_r	N	N_c	T
0.5B fp32	896	24	8	20	128	128	128
0.5B bf16	896	24	8	20	128	128	128
1.5B bf16	1536	28	9	23	128	128	128
7B bf16	3584	28	9	23	128	128	128

Table 3: Measured configurations. Layer indices refer to block outputs.

Verification and controls. Each session supplies bit accuracy, matches and tails for paired and blind modes. We summarize medians and fractions below four thresholds. Prefix checks average $T \in \{8, 16, 32, 64, 128\}$ states. Two wrong keys project the same marked/clean states; each has its own blind calibration. Blind no-mark checks a clean generation with the owner key. Paired no-mark compares that cached clean generation with a full clean teacher-forced replay, isolating numerical cache/full-pass differences. Its threshold rates estimate the empirical FPR of the recommended verifier. Substitution constructs

$$h_t^{\text{sub}} = h_t(x, y_{<t}) + \alpha \|h_t(x, y_{<t})\|_2 \frac{C_K b}{\|C_K b\|_2} \tag{8}$$

at the read layer. It bypasses the intermediate *marked* computation, not the full clean reference replay. It represents known-key fabrication.

5 Results

5.1 Long-horizon fidelity and scale/precision

At fixed 0.5B, fp32 to bf16 lowers identical continuations from 0.688 to 0.258 and paired accuracy from 0.978 to 0.904. Precision dominates this measured contrast; bf16 size effects are smaller and nonmonotonic. The v1 claim of unchanged text does not survive the 128-token test. Conditional

Model/dtype	B	Identical	First div.	D_{KL}	Agreement
fp32					
0.5B fp32	16 phys.	0.688	128	8.30×10^{-5}	0.816
0.5B fp32	64 phys.	0.641	128	8.17×10^{-5}	0.805
bf16					
0.5B bf16	16 phys.	0.258	40.5	7.60×10^{-4}	0.526
0.5B bf16	64 phys.	0.281	43.5	7.57×10^{-4}	0.549
1.5B bf16	16 phys.	0.203	48.5	7.88×10^{-4}	0.526
1.5B bf16	64 phys.	0.227	56	7.88×10^{-4}	0.588
7B bf16	16 phys.	0.211	61	5.47×10^{-4}	0.664
7B bf16	64 phys.	0.250	67	5.27×10^{-4}	0.703

Table 4: Autoregressive fidelity grouped by dtype, separately for each physical payload size. First divergence is zero-indexed and right-censored at the run length; KL is marked-to-clean, through the first differing prediction (inclusive).

KL stays small, yet a single changed greedy decision can alter subsequent context. A nearly tied

decision is a plausible trigger; top-two margins were not measured. The matched 0.5B comparison separates dtype from size and reproduces most of the fidelity loss previously seen against larger bf16 models. At fixed $\alpha = 0.01$, bf16 is a different operating point: rounding and cache/replay numerics can change both fidelity and recovery. This is not a universal precision law.

5.2 Detection and recovery

Model/dtype	B	Mode	Bit acc.	Matches	Nominal median p	$< 10^{-2}$	$< 10^{-3}$	$< 10^{-4}$	$< 10^{-5}$	Full payload
fp32										
0.5B fp32	16 phys.	paired	0.978	16	1.53×10^{-5}	0.984	0.906	0.750	0.000	0.750
0.5B fp32	16 phys.	blind	0.488	8	5.98×10^{-1}	0.008	0.000	0.000	0.000	0.000
0.5B fp32	64 phys.	paired	0.839	54	9.98×10^{-9}	1.000	0.977	0.914	0.859	0.344
0.5B fp32	64 phys.	blind	0.500	32	5.50×10^{-1}	0.008	0.000	0.000	0.000	0.000
0.5B fp32	16 log. (64)	paired	0.928	15	2.59×10^{-4}	0.883	0.688	0.344	0.000	0.344
0.5B fp32	16 log. (64)	blind	0.507	8	5.98×10^{-1}	0.000	0.000	0.000	0.000	0.000
bf16										
0.5B bf16	16 phys.	paired	0.904	15	2.59×10^{-4}	0.805	0.602	0.227	0.000	0.227
0.5B bf16	16 phys.	blind	0.497	8	5.98×10^{-1}	0.000	0.000	0.000	0.000	0.000
0.5B bf16	64 phys.	paired	0.731	47	1.13×10^{-4}	0.906	0.734	0.438	0.273	0.039
0.5B bf16	64 phys.	blind	0.504	33	4.50×10^{-1}	0.008	0.000	0.000	0.000	0.000
0.5B bf16	16 log. (64)	paired	0.822	13	1.06×10^{-2}	0.484	0.172	0.039	0.000	0.039
0.5B bf16	16 log. (64)	blind	0.512	8	5.98×10^{-1}	0.000	0.000	0.000	0.000	0.000
1.5B bf16	16 phys.	paired	0.957	16	1.53×10^{-5}	0.938	0.836	0.547	0.000	0.547
1.5B bf16	16 phys.	blind	0.505	8	5.98×10^{-1}	0.000	0.000	0.000	0.000	0.000
1.5B bf16	64 phys.	paired	0.799	51.5	5.84×10^{-7}	0.984	0.953	0.852	0.695	0.172
1.5B bf16	64 phys.	blind	0.503	32	5.50×10^{-1}	0.000	0.000	0.000	0.000	0.000
1.5B bf16	16 log. (64)	paired	0.896	15	2.59×10^{-4}	0.797	0.531	0.172	0.000	0.172
1.5B bf16	16 log. (64)	blind	0.507	8	5.98×10^{-1}	0.008	0.000	0.000	0.000	0.000
7B bf16	16 phys.	paired	0.914	15	2.59×10^{-4}	0.828	0.617	0.266	0.000	0.266
7B bf16	16 phys.	blind	0.525	8	5.98×10^{-1}	0.008	0.000	0.000	0.000	0.000
7B bf16	64 phys.	paired	0.757	49	1.22×10^{-5}	0.898	0.781	0.617	0.445	0.078
7B bf16	64 phys.	blind	0.502	32	5.50×10^{-1}	0.000	0.000	0.000	0.000	0.000
7B bf16	16 log. (64)	paired	0.850	14	2.09×10^{-3}	0.617	0.336	0.078	0.000	0.078
7B bf16	16 log. (64)	blind	0.490	8	5.98×10^{-1}	0.000	0.000	0.000	0.000	0.000

Table 5: Rows are grouped by dtype. Physical rows report detection over distinct code coordinates; logical rows report sixteen-bit payload recovery after majority vote. Threshold columns are per-session fractions. Full payload is the fraction with sixteen correct logical bits. Logical tails are nominal (payload-randomised): a null majority vote favours +1 with probability 11/16.

Paired recovery also drops when only the 0.5B dtype changes. Its variation across bf16 model sizes is smaller than that matched precision drop and is nonmonotonic; 7B bf16 is not worse than 0.5B bf16. Blind recovery stays near chance. Sixty-four-physical-coordinate tails are substantially smaller than the sixteen-coordinate tails, even though majority-decoded logical accuracy is lower than uncoded accuracy. A fixed normalized perturbation budget is shared among more columns: coding does not automatically improve payload recovery. A stronger detection statistic is not a larger payload.

Figure 1 and Table 6 distinguish physical detection from logical recovery over prefix length. More tokens change noise and accumulated cache effects; they do not increase the number of code-coordinate trials.

5.3 Null controls and substitution

0.5B fp32: the five physical null-control series have rates from 0.000 to 0.023 at 10^{-2}, and from 0.000 to 0.008 at 10^{-3}. All have zero exceedances at 10^{-4} and 10^{-5}. 0.5B bf16: the five physical null-control series have rates from 0.008 to 0.023 at 10^{-2}, and from 0.000 to 0.008 at 10^{-3}. All have zero exceedances at 10^{-4} and 10^{-5}. 1.5B bf16: the five physical null-control series have rates from 0.000 to 0.016 at 10^{-2}, and from 0.000 to 0.000 at 10^{-3}. All have zero exceedances at 10^{-4} and 10^{-5}. 7B bf16: the five physical null-control series have rates from 0.000 to 0.023 at 10^{-2}, and from 0.000 to 0.000 at 10^{-3}. All have zero exceedances at 10^{-4} and 10^{-5}.These rates fluctuate around low null exceedance rates; the occasional rate above nominal does not establish a calibrated bound.

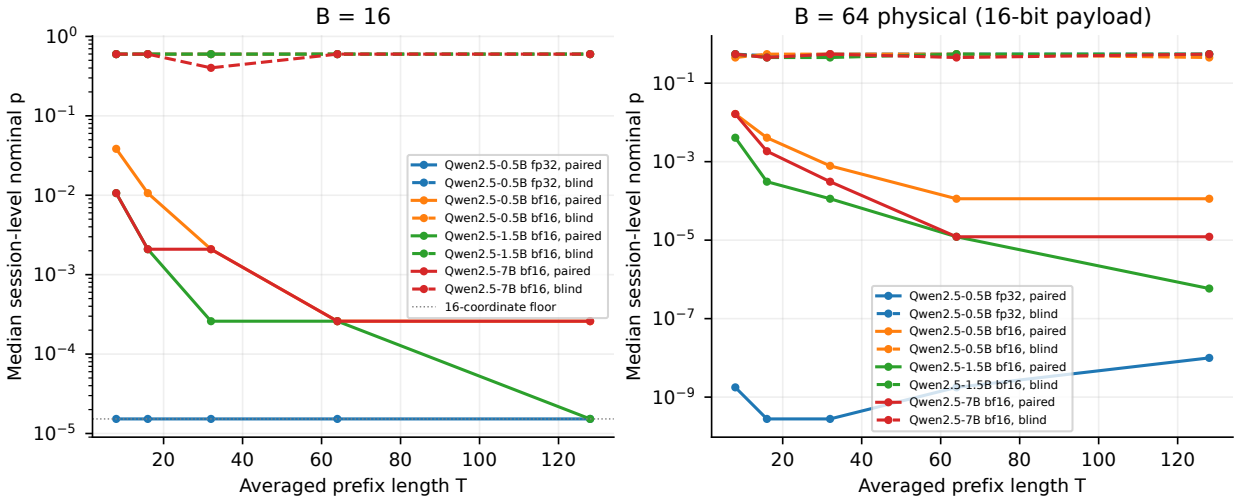


Figure 1: Median session-level nominal detection tails for sixteen and sixty-four physical coordinates. Solid: paired; dashed: blind. The sixteen-coordinate floor applies only to the left panel.

Model/dtype	Coordinates	Mode	$T = 8$	$T = 16$	$T = 32$	$T = 64$	$T = 128$
0.5B fp32	16 phys.	paired	1.53×10^{-5}	1.53×10^{-5}	1.53×10^{-5}	1.53×10^{-5}	1.53×10^{-5}
0.5B fp32	16 phys.	blind	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}
0.5B fp32	64 phys.	paired	1.77×10^{-9}	2.78×10^{-10}	2.78×10^{-10}	1.77×10^{-9}	9.98×10^{-9}
0.5B fp32	64 phys.	blind	5.50×10^{-1}	5.00×10^{-1}	5.50×10^{-1}	5.50×10^{-1}	5.50×10^{-1}
0.5B fp32	16 log. (64)	paired	2.59×10^{-4}	2.59×10^{-4}	2.59×10^{-4}	2.59×10^{-4}	2.59×10^{-4}
0.5B fp32	16 log. (64)	blind	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}
0.5B bf16	16 phys.	paired	3.84×10^{-2}	1.06×10^{-2}	2.09×10^{-3}	2.59×10^{-4}	2.59×10^{-4}
0.5B bf16	16 phys.	blind	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}
0.5B bf16	64 phys.	paired	1.64×10^{-2}	4.07×10^{-3}	7.81×10^{-4}	1.13×10^{-4}	1.13×10^{-4}
0.5B bf16	64 phys.	blind	4.50×10^{-1}	5.50×10^{-1}	5.50×10^{-1}	5.50×10^{-1}	4.50×10^{-1}
0.5B bf16	16 log. (64)	paired	1.05×10^{-1}	3.84×10^{-2}	1.06×10^{-2}	1.06×10^{-2}	1.06×10^{-2}
0.5B bf16	16 log. (64)	blind	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}
1.5B bf16	16 phys.	paired	1.06×10^{-2}	2.09×10^{-3}	2.59×10^{-4}	2.59×10^{-4}	1.53×10^{-5}
1.5B bf16	16 phys.	blind	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}
1.5B bf16	64 phys.	paired	4.07×10^{-3}	3.09×10^{-4}	1.13×10^{-4}	1.22×10^{-5}	5.84×10^{-7}
1.5B bf16	64 phys.	blind	5.50×10^{-1}	4.50×10^{-1}	4.50×10^{-1}	5.50×10^{-1}	5.50×10^{-1}
1.5B bf16	16 log. (64)	paired	3.84×10^{-2}	1.06×10^{-2}	1.06×10^{-2}	2.09×10^{-3}	2.59×10^{-4}
1.5B bf16	16 log. (64)	blind	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}
7B bf16	16 phys.	paired	1.06×10^{-2}	2.09×10^{-3}	2.09×10^{-3}	2.59×10^{-4}	2.59×10^{-4}
7B bf16	16 phys.	blind	5.98×10^{-1}	5.98×10^{-1}	4.02×10^{-1}	5.98×10^{-1}	5.98×10^{-1}
7B bf16	64 phys.	paired	1.64×10^{-2}	1.84×10^{-3}	3.09×10^{-4}	1.22×10^{-5}	1.22×10^{-5}
7B bf16	64 phys.	blind	5.50×10^{-1}	4.50×10^{-1}	5.50×10^{-1}	4.50×10^{-1}	5.50×10^{-1}
7B bf16	16 log. (64)	paired	7.17×10^{-2}	2.45×10^{-2}	1.06×10^{-2}	2.09×10^{-3}	2.09×10^{-3}
7B bf16	16 log. (64)	blind	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}	5.98×10^{-1}

Table 6: Median session-level nominal tails versus prefix length. Physical and logical statistics are separated; logical prefix tails are reconstructed from session records.

Zero observed exceedances cannot validate a 10^{-5} false-positive guarantee with only 128 sessions per series. The physical controls test the null required by Eq. (7); repetition is not a reason to discard those coordinates. Paired no-mark directly measures the empirical FPR of the recommended verifier, including cached-generation versus full-replay numerical differences. 0.5B fp32: B16 physical: 0/128 below 10^{-3} (0 below 10^{-2}); B64 physical: 0/128 below 10^{-3} (2 below 10^{-2}). 0.5B bf16: B16 physical: 0/128 below 10^{-3} (0 below 10^{-2}); B64 physical: 0/128 below 10^{-3} (1 below 10^{-2}). 1.5B bf16: B16 physical: 0/128 below 10^{-3} (0 below 10^{-2}); B64 physical: 0/128 below 10^{-3} (0 below 10^{-2}). 7B bf16: B16 physical: 0/128 below 10^{-3} (0 below 10^{-2}); B64 physical: 0/128 below 10^{-3} (1 below 10^{-2}). Pending paired no-mark conditions: none. Substitution recovers all sixteen uncoded bits

Model/dtype	B	Control/mode	Bit acc.	Matches	Nominal median p	$< 10^{-2}$	$< 10^{-3}$	$< 10^{-4}$	$< 10^{-5}$
0.5B fp32	16 phys.	wrong 1, paired	0.515	8	5.98×10^{-1}	0.008	0.000	0.000	0.000
0.5B fp32	16 phys.	wrong 1, blind	0.488	8	5.98×10^{-1}	0.008	0.000	0.000	0.000
0.5B fp32	16 phys.	wrong 2, paired	0.474	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
0.5B fp32	16 phys.	wrong 2, blind	0.512	8	5.98×10^{-1}	0.008	0.000	0.000	0.000
0.5B fp32	16 phys.	nomark blind	0.491	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
0.5B fp32	16 phys.	substitution paired	1.000	16	1.53×10^{-5}	1.000	1.000	1.000	0.000
0.5B fp32	16 phys.	substitution blind	0.579	9	4.02×10^{-1}	0.023	0.008	0.000	0.000
0.5B fp32	16 phys.	no-mark, paired	0.510	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
0.5B fp32	64 phys.	wrong 1, paired	0.512	33	4.50×10^{-1}	0.008	0.008	0.000	0.000
0.5B fp32	64 phys.	wrong 1, blind	0.506	32	5.50×10^{-1}	0.008	0.000	0.000	0.000
0.5B fp32	64 phys.	wrong 2, paired	0.484	31	6.46×10^{-1}	0.000	0.000	0.000	0.000
0.5B fp32	64 phys.	wrong 2, blind	0.507	32	5.50×10^{-1}	0.023	0.000	0.000	0.000
0.5B fp32	64 phys.	nomark blind	0.500	32	5.50×10^{-1}	0.000	0.000	0.000	0.000
0.5B fp32	64 phys.	substitution paired	1.000	64	5.42×10^{-20}	1.000	1.000	1.000	1.000
0.5B fp32	64 phys.	substitution blind	0.542	35	2.66×10^{-1}	0.055	0.000	0.000	0.000
0.5B fp32	64 phys.	no-mark, paired	0.505	33	4.50×10^{-1}	0.016	0.000	0.000	0.000
0.5B fp32	16 log. (64)	wrong 1, paired	0.523	8	5.98×10^{-1}	0.008	0.000	0.000	0.000
0.5B fp32	16 log. (64)	wrong 1, blind	0.507	8	5.98×10^{-1}	0.008	0.000	0.000	0.000
0.5B fp32	16 log. (64)	wrong 2, paired	0.482	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
0.5B fp32	16 log. (64)	wrong 2, blind	0.522	8	5.98×10^{-1}	0.008	0.000	0.000	0.000
0.5B fp32	16 log. (64)	nomark blind	0.509	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
0.5B fp32	16 log. (64)	substitution paired	1.000	16	1.53×10^{-5}	1.000	1.000	1.000	0.000
0.5B fp32	16 log. (64)	substitution blind	0.566	9	4.02×10^{-1}	0.016	0.000	0.000	0.000
0.5B fp32	16 log. (64)	no-mark, paired	0.504	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
0.5B bf16	16 phys.	wrong 1, paired	0.498	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
0.5B bf16	16 phys.	wrong 1, blind	0.487	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
0.5B bf16	16 phys.	wrong 2, paired	0.467	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
0.5B bf16	16 phys.	wrong 2, blind	0.507	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
0.5B bf16	16 phys.	nomark blind	0.501	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
0.5B bf16	16 phys.	substitution paired	1.000	16	1.53×10^{-5}	1.000	1.000	1.000	0.000
0.5B bf16	16 phys.	substitution blind	0.588	10	2.27×10^{-1}	0.008	0.000	0.000	0.000
0.5B bf16	16 phys.	no-mark, paired	0.495	8	5.98×10^{-1}	0.000	0.000	0.000	0.000

Table 7: Null and substitution controls, including sixty-four-physical-bit detection and sixteen-logical-bit recovery. No-mark threshold columns are empirical FPRs; paired rows appear when measured. All tails are nominal; logical tails are nominal (payload-randomised). Wrong keys are fixed controls.

in every measured model. It matches or exceeds honest paired recovery without carrying the mark through intervening blocks. A near-identity local residual response is a plausible explanation, not a measured Jacobian. The result excludes interpreting successful decoding as proof of execution. Precommitment prevents later rewriting; it cannot make an earlier fabrication truthful.

6 Failure of the tested projected-logit decoder

The legacy `watermark_provenance_result.json` probe used sixteen prompts and eight prediction positions, not eight prompts. At $B = 16$, $\alpha = 0.01$, it reports residual accuracy 0.919, tested logit accuracy 0.568, KL 3.50×10^{-4} and token agreement 1.000. The logit decoder correlates differences against WC_K. Unembedding W is linear; final RMS normalization is nonlinear. Weak recovery by this decoder does not prove all logit or text detectors impossible. No text-only detector is validated here. The broader empirical study is companion Paper B, *Engineered CDMA superposition in pretrained transformers collapses at every untrained non-linearity* [6].

Model/dtype	B	Control/mode	Bit acc.	Matches	Nominal median p	$< 10^{-2}$	$< 10^{-3}$	$< 10^{-4}$	$< 10^{-5}$
0.5B bf16	64 phys.	wrong 1, paired	0.508	33	4.50×10^{-1}	0.008	0.000	0.000	0.000
0.5B bf16	64 phys.	wrong 1, blind	0.500	32	5.50×10^{-1}	0.008	0.000	0.000	0.000
0.5B bf16	64 phys.	wrong 2, paired	0.496	32	5.50×10^{-1}	0.023	0.008	0.000	0.000
0.5B bf16	64 phys.	wrong 2, blind	0.503	32	5.50×10^{-1}	0.008	0.000	0.000	0.000
0.5B bf16	64 phys.	nomark blind	0.501	33	4.50×10^{-1}	0.008	0.000	0.000	0.000
0.5B bf16	64 phys.	substitution paired	1.000	64	5.42×10^{-20}	1.000	1.000	1.000	1.000
0.5B bf16	64 phys.	substitution blind	0.544	35	2.66×10^{-1}	0.039	0.016	0.000	0.000
0.5B bf16	64 phys.	no-mark, paired	0.503	32	5.50×10^{-1}	0.008	0.000	0.000	0.000
0.5B bf16	16 log. (64)	wrong 1, paired	0.518	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
0.5B bf16	16 log. (64)	wrong 1, blind	0.500	8	5.98×10^{-1}	0.008	0.000	0.000	0.000
0.5B bf16	16 log. (64)	wrong 2, paired	0.487	8	5.98×10^{-1}	0.008	0.000	0.000	0.000
0.5B bf16	16 log. (64)	wrong 2, blind	0.521	8	5.98×10^{-1}	0.008	0.000	0.000	0.000
0.5B bf16	16 log. (64)	nomark blind	0.513	8	5.98×10^{-1}	0.008	0.000	0.000	0.000
0.5B bf16	16 log. (64)	substitution paired	1.000	16	1.53×10^{-5}	1.000	1.000	1.000	0.000
0.5B bf16	16 log. (64)	substitution blind	0.564	9	4.02×10^{-1}	0.023	0.000	0.000	0.000
0.5B bf16	16 log. (64)	no-mark, paired	0.504	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
1.5B bf16	16 phys.	wrong 1, paired	0.500	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
1.5B bf16	16 phys.	wrong 1, blind	0.472	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
1.5B bf16	16 phys.	wrong 2, paired	0.476	7.5	6.85×10^{-1}	0.000	0.000	0.000	0.000
1.5B bf16	16 phys.	wrong 2, blind	0.500	8	5.98×10^{-1}	0.016	0.000	0.000	0.000
1.5B bf16	16 phys.	nomark blind	0.483	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
1.5B bf16	16 phys.	substitution paired	1.000	16	1.53×10^{-5}	1.000	1.000	1.000	0.000
1.5B bf16	16 phys.	substitution blind	0.623	10	2.27×10^{-1}	0.023	0.008	0.000	0.000
1.5B bf16	16 phys.	no-mark, paired	0.500	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
1.5B bf16	64 phys.	wrong 1, paired	0.505	32	5.50×10^{-1}	0.000	0.000	0.000	0.000
1.5B bf16	64 phys.	wrong 1, blind	0.496	32	5.50×10^{-1}	0.000	0.000	0.000	0.000
1.5B bf16	64 phys.	wrong 2, paired	0.496	31	6.46×10^{-1}	0.016	0.000	0.000	0.000
1.5B bf16	64 phys.	wrong 2, blind	0.493	31	6.46×10^{-1}	0.008	0.000	0.000	0.000
1.5B bf16	64 phys.	nomark blind	0.501	32	5.50×10^{-1}	0.000	0.000	0.000	0.000
1.5B bf16	64 phys.	substitution paired	1.000	64	5.42×10^{-20}	1.000	1.000	1.000	1.000
1.5B bf16	64 phys.	substitution blind	0.565	36	1.91×10^{-1}	0.125	0.031	0.000	0.000
1.5B bf16	64 phys.	no-mark, paired	0.499	32	5.50×10^{-1}	0.000	0.000	0.000	0.000

Table 8: Null and substitution controls, including sixty-four-physical-bit detection and sixteen-logical-bit recovery. No-mark threshold columns are empirical FPRs; paired rows appear when measured. All tails are nominal; logical tails are nominal (payload-randomised). Wrong keys are fixed controls. Continued.

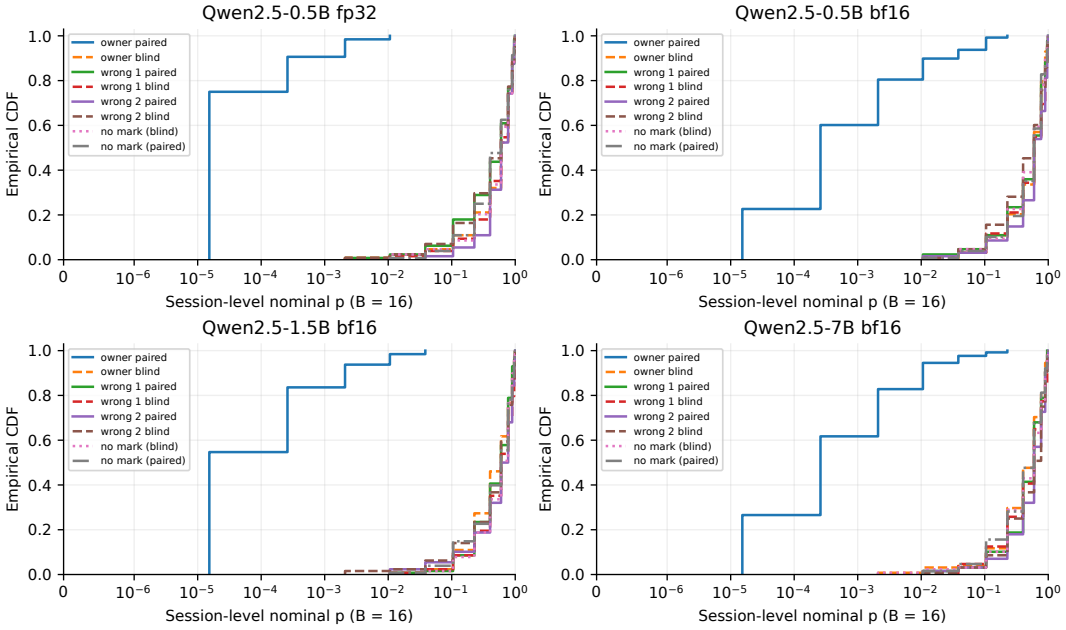


Figure 2: Sixteen-physical-coordinate nominal-tail ECDFs: owner and two wrong keys, paired and blind; no-mark is blind and, when measured, paired.

Model/dtype	B	Control/mode	Bit acc.	Matches	Nominal median p	$< 10^{-2}$	$< 10^{-3}$	$< 10^{-4}$	$< 10^{-5}$
1.5B bf16	16 log. (64)	wrong 1, paired	0.505	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
1.5B bf16	16 log. (64)	wrong 1, blind	0.507	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
1.5B bf16	16 log. (64)	wrong 2, paired	0.500	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
1.5B bf16	16 log. (64)	wrong 2, blind	0.492	8	5.98×10^{-1}	0.008	0.000	0.000	0.000
1.5B bf16	16 log. (64)	nomark blind	0.500	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
1.5B bf16	16 log. (64)	substitution paired	1.000	16	1.53×10^{-5}	1.000	1.000	1.000	0.000
1.5B bf16	16 log. (64)	substitution blind	0.597	10	2.27×10^{-1}	0.023	0.008	0.000	0.000
1.5B bf16	16 log. (64)	no-mark, paired	0.485	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
7B bf16	16 phys.	wrong 1, paired	0.508	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
7B bf16	16 phys.	wrong 1, blind	0.504	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
7B bf16	16 phys.	wrong 2, paired	0.483	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
7B bf16	16 phys.	wrong 2, blind	0.493	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
7B bf16	16 phys.	nomark blind	0.512	8	5.98×10^{-1}	0.008	0.000	0.000	0.000
7B bf16	16 phys.	substitution paired	1.000	16	1.53×10^{-5}	1.000	1.000	1.000	0.000
7B bf16	16 phys.	substitution blind	0.767	12	3.84×10^{-2}	0.312	0.148	0.031	0.000
7B bf16	16 phys.	no-mark, paired	0.523	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
7B bf16	64 phys.	wrong 1, paired	0.511	33	4.50×10^{-1}	0.023	0.000	0.000	0.000
7B bf16	64 phys.	wrong 1, blind	0.494	31	6.46×10^{-1}	0.008	0.000	0.000	0.000
7B bf16	64 phys.	wrong 2, paired	0.493	32	5.50×10^{-1}	0.000	0.000	0.000	0.000
7B bf16	64 phys.	wrong 2, blind	0.498	32	5.50×10^{-1}	0.000	0.000	0.000	0.000
7B bf16	64 phys.	nomark blind	0.499	32	5.50×10^{-1}	0.000	0.000	0.000	0.000
7B bf16	64 phys.	substitution paired	1.000	64	5.42×10^{-20}	1.000	1.000	1.000	1.000
7B bf16	64 phys.	substitution blind	0.618	39	5.17×10^{-2}	0.375	0.172	0.031	0.008
7B bf16	64 phys.	no-mark, paired	0.507	32	5.50×10^{-1}	0.008	0.000	0.000	0.000
7B bf16	16 log. (64)	wrong 1, paired	0.520	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
7B bf16	16 log. (64)	wrong 1, blind	0.491	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
7B bf16	16 log. (64)	wrong 2, paired	0.507	8	5.98×10^{-1}	0.008	0.000	0.000	0.000
7B bf16	16 log. (64)	wrong 2, blind	0.488	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
7B bf16	16 log. (64)	nomark blind	0.486	8	5.98×10^{-1}	0.000	0.000	0.000	0.000
7B bf16	16 log. (64)	substitution paired	1.000	16	1.53×10^{-5}	1.000	1.000	1.000	0.000
7B bf16	16 log. (64)	substitution blind	0.662	11	1.05×10^{-1}	0.062	0.016	0.000	0.000
7B bf16	16 log. (64)	no-mark, paired	0.515	8	5.98×10^{-1}	0.008	0.008	0.000	0.000

Table 9: Null and substitution controls, including sixty-four-physical-bit detection and sixteen-logical-bit recovery. No-mark threshold columns are empirical FPRs; paired rows appear when measured. All tails are nominal; logical tails are nominal (payload-randomised). Wrong keys are fixed controls. Continued.

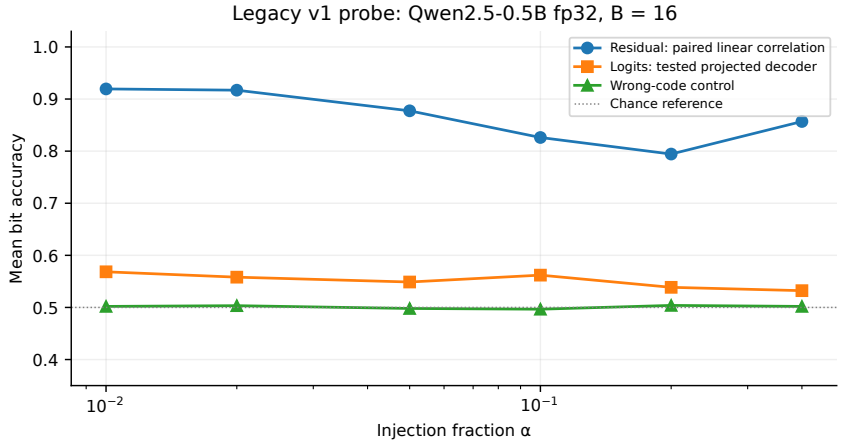


Figure 3: Legacy residual/logit recovery versus amplitude; not a long-rollout or blind-verification test.

7 Related work

Kirchenbauer et al. [3] favor pseudorandom green tokens and detect from text without model parameters. Our activation-log tag has a different access model and does not inherit that black-box guarantee. SEAL [1] edits latent subspaces for model-ownership watermarking with white-/black-box settings. EverTracer [8] uses fine-tuning and calibrated probability variation for probabilistic model fingerprinting. Both concern model ownership, not per-session audit labels. No robustness comparison is measured here. Certificate Transparency [4] supplies append-only commitment machinery, not truthfulness of logged statements. zkLLM [5] instead constructs zero-knowledge machinery to prove LLM computation. Its soundness target differs from correlation with a secret activation code; activation tagging is not a lightweight equivalent of verifiable inference.

8 Limitations

The experiments cover one prompt source, greedy decoding, one placement rule and fixed public research seeds. They do not establish semantic fidelity, sampling quality, paraphrase robustness or stability under fine-tuning. Model, tokenizer, precision or kernel drift can break paired replay. The matched 0.5B experiment shows that precision, rather than size, dominates the measured degradation; fixed $\alpha = 0.01$ is not precision-invariant. This attribution is limited to these models and kernels, without additional amplitudes or repeated runs. Only two wrong keys are tested; physical null independence remains an assumption checked imperfectly by controls. Logical majority votes have tie bias, and sixteen payload bits are not a unique session identifier. Neither detection nor failure to detect proves origin or non-origin against a malicious provider.

Capture integrity, logging completeness, retention, commitment privacy and production costs require separate evaluation. Compact projections are not zero-knowledge disclosures. Paired verification needs text privately. Signed unmodified activation logs also support internal replay consistency without perturbing generation. The incremental utility of the perturbation over that baseline remains unproven; these experiments characterize the mechanism and its negative results rather than establish a deployment advantage.

Reproducibility

Repository: `https://github.com/ttzrs/neural-cdma`. Previous publication: 10.5281/zenodo.20570543, a concept DOI resolving to v1 record 20570544 [7]. Neither identifies this revision. `watermark_scale.py` defines the current experiment; `watermark_provenance.py` supports the legacy probe. Table 10 identifies exact model revisions. New JSONs record `config.model_revision`; the earlier snapshots are listed explicitly. The experiment saves decoding statistics, not the committed tuple in Eq. (2). Publishing reproducible committed tuples remains future work. Run `bash build.sh` in `papers/forensic/` to regenerate tables, figures,

Model/dtype	Hugging Face model revision	Provenance
0.5B fp32	`060db6499f32faf8b98477b0a26969ef7d8b9987`	legacy snapshot
0.5B bf16	`060db6499f32faf8b98477b0a26969ef7d8b9987`	legacy snapshot
1.5B bf16	`8faed761d45a263340a0528343f099c05c9a4323`	legacy snapshot
7B bf16	`d149729398750b98c0af14eb82c78cfe92750796`	legacy snapshot

Table 10: Model revisions. Legacy snapshots identify the earlier runs whose JSONs omitted this field; new JSON revisions take precedence.

BibTeX references and PDF. Numeric claims use generated macros; full source hashes are in `tables/source_manifest.json`. The package's demonstrative reinjection verifier is distinct from the committed-record protocol specified here.

References

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[2] Simon Josefsson and Ilari Liusvaara. Edwards-curve digital signature algorithm (EdDSA). RFC 8032, Internet Research Task Force, January 2017. URL https://www.rfc-editor.org/rfc/rfc8032.

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[4] Ben Laurie, Adam Langley, and Emilia Kasper. Certificate transparency. RFC 6962, Internet Engineering Task Force, June 2013. URL https://www.rfc-editor.org/rfc/rfc6962.

[5] Haochen Sun, Jason Li, and Hongyang Zhang. zkLLM: Zero knowledge proofs for large language models, 2024. URL https://arxiv.org/abs/2404.16109. Accepted to ACM CCS 2024; cited arXiv author version.

[6] Jose Joaquín Vargas Altalaguerri. Engineered CDMA superposition in pretrained transformers collapses at every untrained non-linearity. Companion manuscript, Paper B, in the same research repository; no external identifier assigned, 2026. URL https://github.com/ttzrs/neural-cdma.

[7] Jose Joaquín Vargas Altalaguerri. Forensic provenance for LLM deployments: Real-time activation watermarking for legal non-repudiation, 2026. URL https://doi.org/10.5281/zenodo.20570543. Previous version, June 2026. Concept DOI resolves to v1 record 10.5281/zenodo.20570544; the non-repudiation claims are corrected here.

[8] Zhenhua Xu, Meng Han, and Wenpeng Xing. EverTracer: Hunting stolen large language models via stealthy and robust probabilistic fingerprint. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 7008–7031. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.emnlp-main.358. URL https://aclanthology.org/2025.emnlp-main.358/.

A Changes from version 1

Issue	v1	This revision
Prompts	"8"; actually 16×8 positions	Separate session/calibration counts
Protocol	Static formula, paired code	Separate paired and blind definitions
Statistics	Pooled deltas	Per-session physical detection; logical recovery
Fidelity/claim	Short agreement; non-repudiation	Long divergence; committed-record consistency

Table 11: Changes from the previous version.